RPractical_esmedullar

Sys.Date()

Part A A1

```
v <- seq(from = 5,to =50, by = 5)
print(length(v))
```

```
## [1] 10
```

```
print(sum(v))
```

```
## [1] 275
```

```
print(mean(v))
```

```
## [1] 27.5
```

A2

```
x <- c(1, 2, 3)
w <- rep(x, times = 4)
z <- rep(x, each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
17 / 5
```

```
## [1] 3.4
```

```
17 %/% 5
```

```
## [1] 3
```

```
2 ^ 10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```
print(pi, digits = 4)
```

```
## [1] 3.142
```

A4

```
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
```

```r
a + b
```

```
## [1]  3  7 11 15
```

```r
a * b
```

```
## [1]  2 12 30 56
```

```r
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a * b)
```

```
## [1] 100
```

A5

```r
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
```

B2

```r
print(students)
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
```

```
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B3

```
students$average <- round((students$midterm + students$final) / 2, 1)
```

B4

```
students$passed <- (students$average >= 70) & (students$attendance >= 75)
```

B5

```
students$standing <- ifelse(students$average >= 90, "honors",
                            ifelse(students$average >= 75, "good", "needs support"))
```

B6

Part C C1

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- (new_students$average >= 70) & (new_students$attendance >= 75)
new_students$standing <- ifelse(new_students$average >= 90, "honors",
                                ifelse(new_students$average >= 75, "good", "needs support"))
students_all <- rbind(students, new_students)
```

C2

```
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
class(students_all)
```

```
## [1] "data.frame"
```

C3

```
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
```

C4

```
bad <- data.frame(ID = 11, FullName = "Zoe", Grade = 88)
tryCatch(
  rbind(students, bad),
  error = function(e) {
    message("rbind failed: ", conditionMessage(e))
```

```
    NULL
  }
)
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```